

Computer Programming

Lab report 7

Lab title

One-Dimensional Array in C++

1. Objective

- To learn the concept of arrays in C++
- To understand declaration and initialization of arrays
- To access elements of arrays
- To copy arrays using loops
- To understand character arrays (strings)

2. Introduction

- An array is a data structure that stores multiple values of the same data type in contiguous memory locations. Arrays are useful when we need to store and process a large amount of similar data.

The first element of an array starts from index 0.

3. Declaration of Array

- The syntax for declaring an array is:

```
dataType arrayName[arraySize];
```

Example

```
int age[4];
```

Here:

- int is the data type
- age is the array name
- 4 is the size of array

4. Initialization of Array

- Arrays can be initialized during declaration.

Example

```
int age[4] = {23, 34, 64, 75};
```

5. Program for Array Initialization

```
#include <iostream>
using namespace std;
```

```
int main()
{
    int age[4];

    age[0] = 23;
    age[1] = 34;
    age[2] = 64;
    age[3] = 75;

    return 0;
}
```

6. Accessing Elements of Array

➤ Array elements are accessed using indexes.

Example

```
age[0];
age[1];
age[2];
age[3];
```

Example of Accessing Value

```
a = age[2];
```

7. Program for Copying Arrays

```
#include <iostream>
using namespace std;
```

```
int main()
{
    int age[4];
    int same_age[4];
    int i = 0;

    age[0] = 23;
    age[1] = 34;
    age[2] = 64;
    age[3] = 75;

    for(i = 0; i < 4; i++)
    {
        same_age[i] = age[i];
    }

    for(i = 0; i < 4; i++)
    {
        cout << same_age[i] << "\t";
    }

    return 0;
}
```

8. Result

- All programs were compiled and executed successfully.

Task 1: Array Declaration and Initialization

Problem Statement

- Declare an integer array called numbers with a size of 5.
Initialize the array with values 10, 20, 30, 40, 50 and display all elements using a loop.

Program

```
#include <iostream>
using namespace std;

int main()
{
    // Declare and initialize array
    int numbers[5] = {10, 20, 30, 40, 50};

    // Display array elements using loop
    for (int i = 0; i < 5; i++)
    {
        cout << numbers[i] << endl;
    }

    return 0;
}
```

Output

```
10
20
30
40
50
```

Task 2: Accessing Array Elements

Problem Statement

- Declare a character array called message to store a word.
Use a loop to display each character on a separate line.

Program

```
#include <iostream>
using namespace std;

int main()
{
    // Declare character array
    char message[] = "HELLO";

    // Display each character
    for (int i = 0; message[i] != '\0'; i++)
```

```

    {
        cout << message[i] << endl;
    }

    return 0;
}

```

Output

H
E
L
L
O

Task 3: Copying Arrays

Problem Statement

- Declare two integer arrays named source and destination.
Copy elements from source array to destination array using a loop.

Program

```

#include <iostream>
using namespace std;

int main()
{
    // Declare arrays
    int source[5] = {5, 10, 15, 20, 25};
    int destination[5];

    // Copy elements from source to destination
    for (int i = 0; i < 5; i++)
    {
        destination[i] = source[i];
    }

    // Display source array
    cout << "Source Array: ";

    for (int i = 0; i < 5; i++)
    {
        cout << source[i] << " ";
    }

    cout << endl;

    // Display destination array
    cout << "Destination Array: ";

    for (int i = 0; i < 5; i++)
    {

```

```

        cout << destination[i] << " ";
    }

    return 0;
}

```

Output

Source Array: 5 10 15 20 25

Destination Array: 5 10 15 20 25

Task 4: Array Sum

Problem Statement

- Write a C++ program that asks the user to enter 5 numbers. Store the numbers in an array and calculate their sum.

Program

```

#include <iostream>
using namespace std;

int main()
{
    int numbers[5];
    int sum = 0;

    // Input numbers from user
    cout << "Enter 5 numbers:" << endl;

    for (int i = 0; i < 5; i++)
    {
        cin >> numbers[i];
    }

    // Calculate sum
    for (int i = 0; i < 5; i++)
    {
        sum = sum + numbers[i];
    }

    // Display sum
    cout << "Sum = " << sum << endl;

    return 0;
}

```

Sample Output

Enter 5 numbers:

```

2
4
6
8
10

```

Sum = 30
